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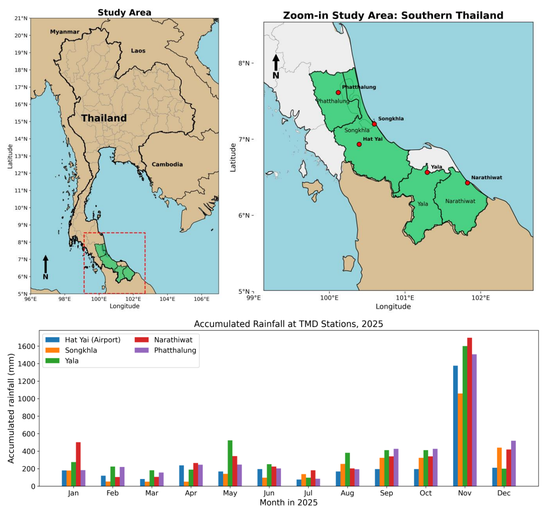
Climate Projections of Monthly-based Extreme Indices for Thailand and Its Comparison During November 2025 Severe Flood in Lower Southern Part

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Introduction

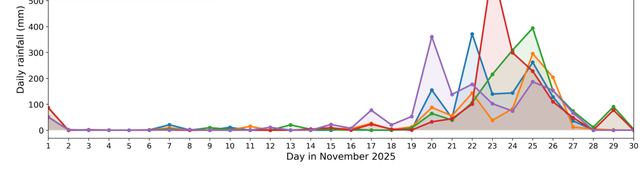
Precipitation extreme indices are traditionally designed for long-term climate assessment and are typically evaluated on annual time scales. In this study, these indices were adapted to a monthly time scale to investigate the unprecedented flooding event that occurred in Lower Southern Thailand during November 2025. Five precipitation extreme indices (mPRCPTOT, mCWD, mR20mm, mR50mm, and mRx5day) were calculated from station observations and compared with CMIP6 ensemble projections under SSP2-4.5, SSP3-7.0, and SSP5-8.5. By bridging climate-scale indicators with event-scale analysis, this study explores the applicability of CMIP6 projections for evaluating extreme rainfall hazards and supporting climate-informed flood risk management. The findings provide insights into how future climate projections can be interpreted for rare but high-impact flood events where observational evidence for future conditions is inherently unavailable.



November 2025 Rainfall Event in Lower Southern Thailand

Daily rainfall data from weather stations reveal that the flood was associated with a period of extreme rainfall between 20 and 26 November rather than continuous rainfall throughout the month. Narathiwat recorded the highest rainfall at approximately 620 mm/day, while Yala and Hat Yai recorded maximums of approximately 400 and 370 mm/day, respectively. This reflects the characteristics of an extreme rainfall event that are significant factors contributing to major flooding in southern Thailand.





Data Sources

1. Daily observation data were obtained from (1) the Global Surface Summary of the Day (GSOD) dataset for 1981–2014 and (2) the Thai Meteorological Department (TMD) for 2025.
2. The gridded data; Climate Hazards Center Infrared Precipitation with Stations (CHIRPS v2) with 0.25 deg resolution.
3. The NEX-GDDP-CMIP6 dataset (0.25 deg) of 34 GCMs from NASA.

Monthly Extreme Precipitation Indices

$$mPRCPTOT = \sum_{d=1}^{D} P_d \cdot I(P_d \geq 1) \qquad mR20mm = \sum_{d=1}^{D} I(P_d \geq 20)$$

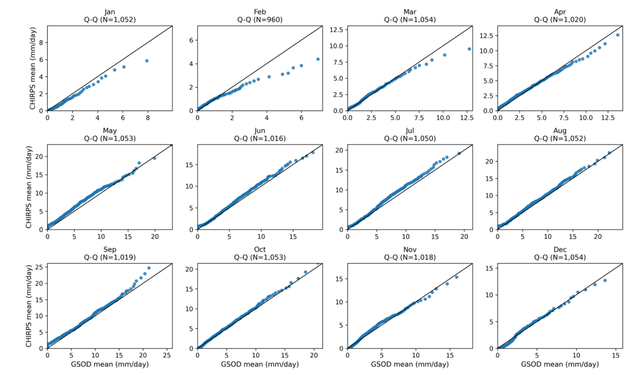
$$mR50mm = \sum_{d=1}^{D} I(P_d \geq 50) \qquad mRx5d = \sum_{i=1}^{5} P_{d+i}$$

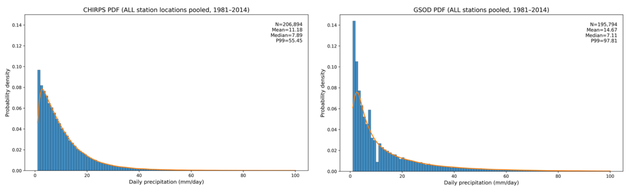
$$mCWD = \max(\text{length of consecutive days where } P_d \geq 1)$$

where P_d is daily precipitation (mm/day), d is the day of the month, D is the total number of days in the month, I is an indicator function equal to 1 for wet days ($P_d \geq 1$)and 0 otherwise.

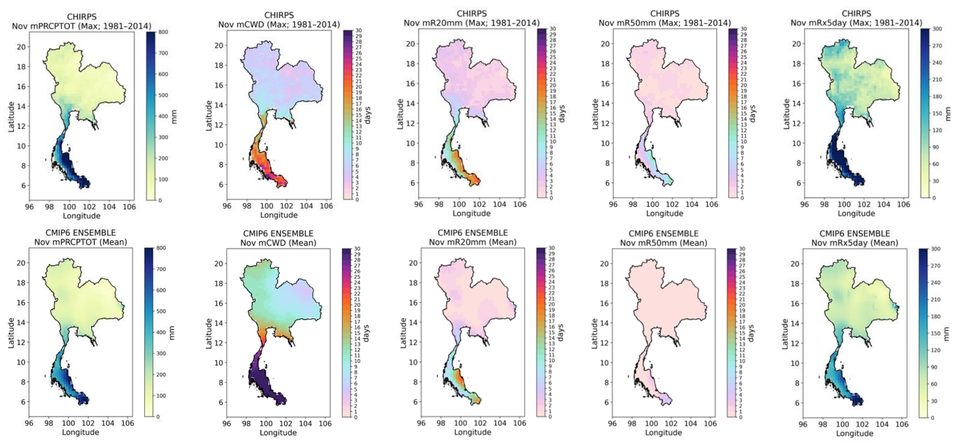
Historical Evaluation of CHIRPS Against Observations

Results from monthly Q-Q plots of daily data are consistent with observed data. Most of the data points are aligned near the 1:1 ratio, which reflects the good ability in representing rainfall amounts. Nevertheless, the differences between the two data sets increase in the high quantiles. CHIRPS shows higher rainfall amounts than GSOD during the late rainy season (September) and lower in January to March. A comparison of the Probability Density Function (PDF) of daily rainfall between CHIRPS and GSOD datasets from 1981–2014 shows that both datasets exhibit similar right-skew distributions and very similar median values (7.89 and 7.11 mm/day, respectively). Differences between the datasets become more apparent at the upper tail of the distribution. The GSOD has significantly higher mean and P99 values than CHIRPS (mean = 14.67 vs. 11.18 mm/day and P99 = 97.81 vs. 55.45 mm/day). This indicates that CHIRPS tends to attenuate the severity of extreme rainfall. Therefore, CHIRPS has been used for analyzing rainfall patterns and conducting long-term climate studies in Thailand.



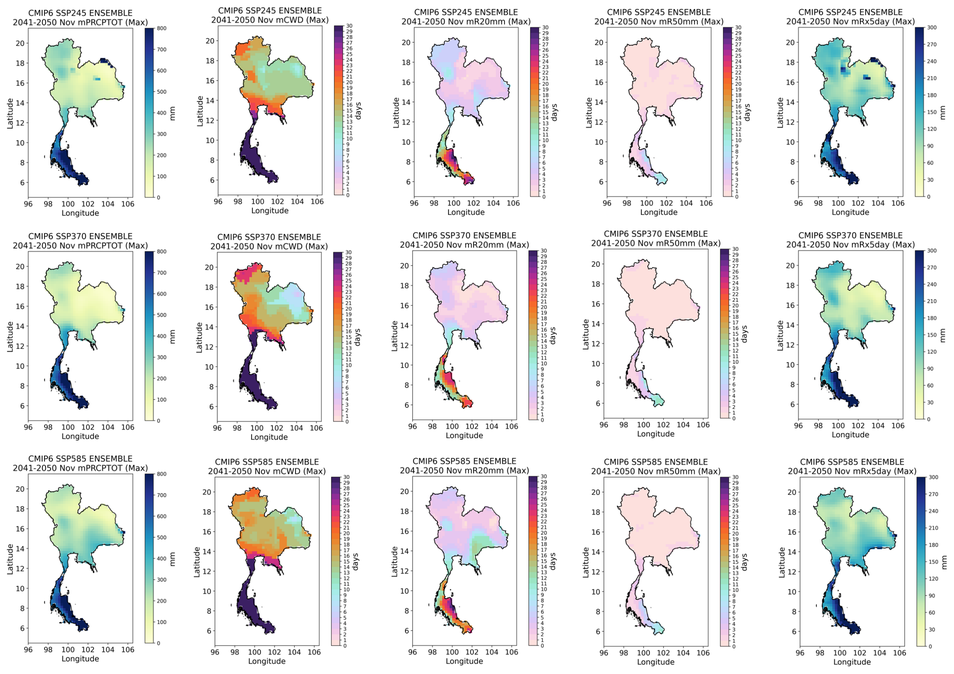


Historical and Future Extreme Precipitation Indices



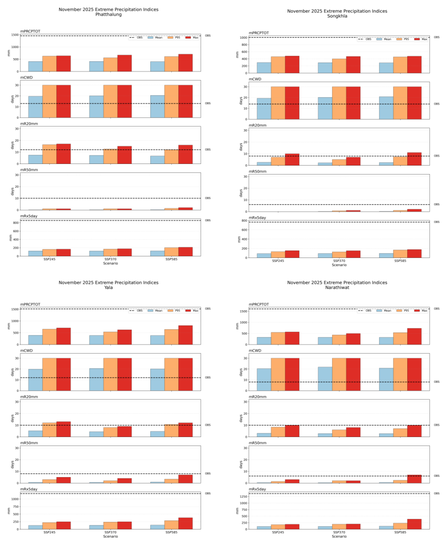
The comparison between CHIRPS observations and the CMIP6 ensemble during November for the period 1981–2014 indicates that the climate models are generally capable of reproducing the spatial distribution of precipitation extremes across Thailand. The highest values of mPRCPTOT and mRx5day are consistently located in the Lower Southern Thailand, reflecting the strong influence of the Northeast Monsoon and moisture transport from the Gulf of Thailand. The CMIP6 ensemble successfully captures these large-scale patterns, although local maxima are generally smoother and less intense than those observed by CHIRPS. For frequency-based indices, the ensemble reproduces the occurrence of heavy rainfall days (mR20mm) reasonably well but tends to overestimate the persistence of wet conditions (mCWD) while underestimating very heavy rainfall events (mR50mm). Overall, the results suggest that the CMIP6 ensemble provides a realistic representation of the spatial characteristics of November precipitation extremes over Thailand and can serve as a useful basis for assessing future changes in flood-related climate hazards.

- Southern Thailand remains the dominant hotspot of November precipitation extremes under all SSPs.
- mPRCPTOT and mRx5day show increasing intensity from SSP2-4.5 to SSP5-8.5.
- mCWD remains very high over the southern peninsula, indicating persistent wet conditions.
- mR20mm becomes more widespread under higher-emission scenarios.
- Future climate conditions may increase the likelihood of flood-producing rainfall events similar to the November 2025 flood in Southern Thailand.



Comparison of November 2025 Extreme Indices

The November 2025 flood event in Lower Southern Thailand exceeded the CMIP6 ensemble mean across all precipitation extreme indices, indicating that the event was associated with a low-probability but high-impact rainfall episode. Although CMIP6 generally underestimated the observed magnitude of mPRCPTOT and mRx5day, the upper-tail realizations (P95 and maximum values) were able to partially represent the severity of heavy rainfall occurrences. These results suggest that ensemble mean projections alone may underestimate future flood hazards. For climate adaptation and disaster risk management, future assessments should incorporate both ensemble mean projections for long-term climate trends and upper-tail (P95/Maximum) realizations for plausible worst-case scenarios. Such an approach can support more robust planning in regions vulnerable to extreme rainfall and flooding where future observations are inherently unavailable.



Conclusion

CMIP6 projections successfully reproduce the overall spatial and temporal characteristics of November precipitation extremes in southern Thailand. However, the severe flood event observed in November 2025 exceeded the CMIP6 ensemble mean at most stations, particularly for mPRCPTOT, mRx5day, and heavy rainfall frequency indices. Extreme realizations represented by the P95 and maximum values were generally closer to the observed conditions than the ensemble mean, indicating that low-probability but high-impact rainfall events may not be fully represented by average climate projections. These findings suggest that future flood risk assessments and climate adaptation planning should consider both ensemble mean conditions and plausible extreme realizations to better capture the range of potential hydroclimatic risks under different climate change scenarios. Beyond informing flood hazard assessment, the methodology provides an evidence base for future interdisciplinary studies examining how climate-induced extreme rainfall may influence human mobility and displacement in vulnerable regions of Thailand and Southeast Asia.

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